EXPERIMENT - 2.

Aim:

To convert the ER diagram of an Indian E-commerce platform into a relational schema and implement it using MySQL with primary keys, foreign keys, NOT NULL, UNIQUE, ON DELETE CASCADE and ON DELETE SET NULL contraints. To insert sample data and demonstrate referential integrity violations.

Source Code:

```
CREATE DATABASE Ecommerce DB;
   USE Ecommerce DB;
CREATE TABLE Customer C
     Customer-ID INT PRIMARY KEY,
      First_NAME VARCHAR (30) NOT NULL,
      Middle_NAME VARCHAR (30),
      Last_Name VARCHAR (30) NOT NULL,
      Email VARCHAR (100) NOT NULL UNIQUE,
      Phone VARCHAR (15) NOT NULL UNIQUE,
       House VARCHAR (50),
        Pin VARCHAR (10),
        City VARCHAR (50)
    );
```

```
CREATE TABLE Seller (
    Seller_ID INT PRIMARY KEY,
    Seller_Name VARCHAR (100) NOT NULL,
    Email VARCHAR (100) NOT NULL UNIQUE,
    Phone VARCHAR (15) NOT NULL UNIQUE
);

CREATE TABLE Categgary (
    Categgary_ID INT PRIMARY KEY,
    Categgary_NAME VARCHAR (50) NOT NULLUNIQUE,
    Desciption VARCHAR (200)
);

CREATE TABLE Product (
    Product_ID INT PRIMARY KEY,
    Name VARCHAR (100) NOT NULL,
    Price DECIMAL (10, 2) NOT NULL,
    Stock INT NOT NULL,
    categgary_ID INT,
    Seller_ID INT,

    FOREIGN KEY (Categgary ID)
    REFERENCE Categgary (Categgary_ID)
    ON DELEDE SET NULL,
```

```
    FOREIGN KEY (Seller_ID)
      REFERENCES Seller (Seller_ID)
       ON DELETE SET NULL
  );

CREATE TABLE Orders (
  Order_ID INT PRIMARY KEY,
  Customer_ID INT NOT NULL,
  Date DATE NOT NULL,
  Amount DECIMAL (10,2) NOT NULL,

  FOREIGN KEY (Customer_ID)
    REFERENCES Customer (Customer_ID)
     ON DELETE CASCADE
    );

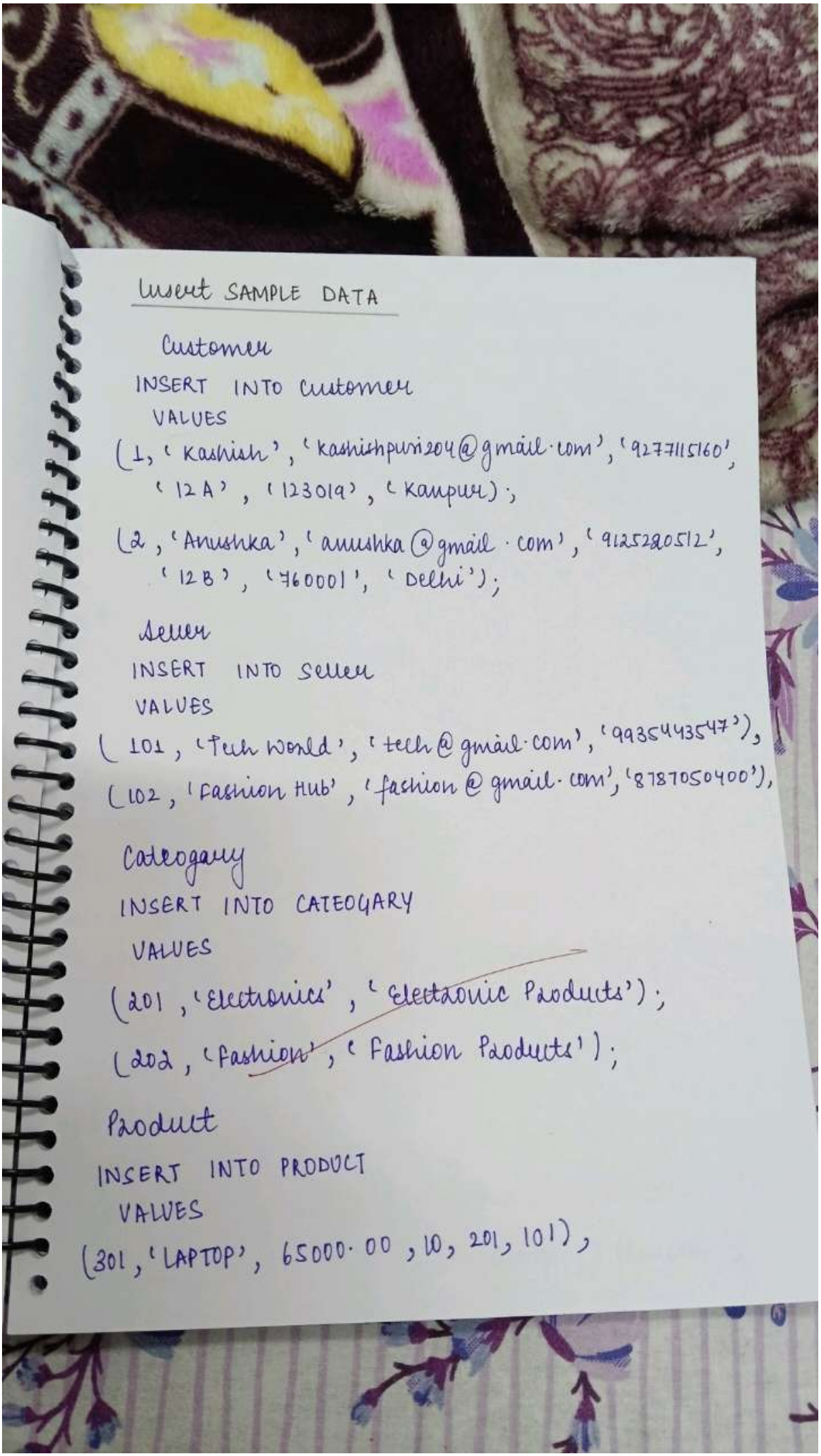
CREATE TABLE OrderItem (
    Order_ID INT,
    Product_ID INT,
    PRIMARY KEY (Order_ID, Product_ID),
    FOREIGN KEY (Order_ID)
    REFRENCES Orders (Order_ID)
     ON DELETE CASCADE,
    FOREIGN KEY (Product_ID)
    REFRENCE Product (Product_ID)
     ON DELETE CASCADE );
```

```
CREATE TABLE Payment (
    Payment_ID INT PRIMARY KEY,
    Order_ID INT NOT NULL UNIQUE,
    Amount DECIMAL (10,2) NOT NULL,
    Method VARCHAR (30) NOT NULL,
    Status VARCHAR (30) NOT NULL,

    FOREIGN KEY (Order_ID)
        REFRENCES Orders (Order_ID)
        ON DELETE CASCADE
    );

CREATE TABLE Delivery (
    Delivery_ID INT PRIMARY KEY,
    Order_ID INT NOT NULL UNIQUE,
    Delivery_Date DATE,
    Status VARCHAR (30) NOT NULL,
    Tracking VARCHAR (30) UNIQUE,

    FOREIGN KEY (Order_ID)
        REFRENCES Orders (Order_ID)
        ON DELETE CASCADE
    );
```



Insert SAMPLE DATA

Customer

```
INSERT INTO Customer
 VALUES
(1, 'Kashish', 'kashishpuri204@gmail.com', '9277115160',
 '12A', '123019', 'Kanpur');

(2, 'Anushka', 'anushka@gmail.com', '9125280512',
 '12B', '760001', 'Delhi');
```

Seller

```
INSERT INTO Seller
 VALUES
(101, 'Tech World', 'tech@gmail.com', '9935443547'),
(102, 'Fashion Hub', 'fashion@gmail.com', '8787050400'),
```

Categgary

```
INSERT INTO CATEOGARY
 VALUES
(201, 'Electronics', 'Electronic Products');
(202, 'Fashion', 'Fashion Products');
```

Product

```
INSERT INTO PRODUCT
 VALUES
(301, 'LAPTOP', 65000.00, 10, 201, 101),
```

(302, 'SM Smartphone', 90000.00, 20, 201, 101);
(303, 'Jacket', '25000.00', 15, 202, 102);

Orders

INSERT INTO Orders
VALUES
(401, 1, '2026-08-19', 65000.00),
(402, 2, '2026-08-19', 30000.00);

Order Item

INSERT INTO Order Item
VALUES
(401, 301),
(402, 302);

Payment

INSERT INTO Payment
VALUES
(501, 401, 65000.00, 'UPI', 'Successful'),
(502, 402, 30000.00, 'Card', 'Successful');

Delivery

INSERT INTO DELIVERY
Values
(601, 401, '2026-08-22'), 'In Transit', 'TRK10001')
(602, 402, '2026-08-23'), 'Processing', 'TRK10002')

Demonstrate of Referential integrity violation

1) Foreign Key violation

Trying to insert an order for a customer who does not exist.

```
INSERT INTO Orders
VALUES
(403, 999, '2026-08-19', 5000.00)
```

Output

```
ERROR 1452 (23000):
cannot add or update a child row:
  a foreign key constraint fails.
```

Here, Customer-ID = 999 doesn't exist in the customer table.

2) PRIMARY Key violation

```
INSERT INTO CUSTOMER
VALUES
(3, 'Rahul', 'rahul@gmail.com', '98765932l', '20A',
'11002', 'Delhi');
```

Output

```
ERROR 1062 (23000):
Duplicate entry 'kashish@gmail.com' for key
'Email' Because the email address is declared
  as UNIQUE'.
```

Unique Constraint Violation

```
INSERT INTO Customer
  VALUES
(3, 'Rahul', 'rahul@gmail.com', '9277115160',
  '20A', '110002', 'Delhi');
```

Output:

```
ERROR 1062 (23000):
```

Duplicate entry 'kashish@gmail.com' for key 'email' Because the email address is declared as UNIQUE.

Demonstration of ON DELETE CASCADE :-

First, delete Customer 1:

```
DELETE FROM Customer
  WHERE Customer_ID = 1;
```

Because orders Customer_ID has ON DELETE CASCADE, all orders belonging to customer 1 will also be deleted.

The related Order Item, Payment & Delivery records will also be deleted through their cascading foreign keys.

Demonstrate of ON DELETE SET NULL:

Delete a seller:

```
Delete FROM ~~SELER~~ seller
WHERE Seller_ID = 101:
```

Since Product Seller_ID uses on DELETE SET NULL, the seller information will be removed but the products will remain.

Check the Product table:

```
SELECT Product_ID, name, Seller_ID
FROM Product.
```

Output:

Product_ID	Name	Seller_ID
301	Laptop	NULL
302	Smartphone	NULL
303	Jacket	102

Relational SCHEMA

```
Customer (
    Customer_ID PK,
    Name,
    Email UNIQUE,
```

Plane UNIQUE ,
House ,
Pin ,
City
)

Seller (
Seller_ID PK,
Seller_Name ,
Email UNIQUE ,
Plane UNIQUE ,
)

Categogary (
Categogary_ID PK ,
Categogary_Name UNIQUE ,
)

Description

Product (
Product_ID PK ,
name ,
Price ,
Stock ,
Categogary_ID FK ,
Seller_ID FK ,
)

Orders (

Order_ID PK ,
Customer_ID PK,
Date ,
Amount .

)

Order Item (

Order_ID PK, FK ,
Product_ID PK, FK .

)

Payment (

Payment_ID PK ,
Order_ID FK UNIQUE ,
Amount ,
Method,
Status

)

Delivery (

Delivery_ID PK ,
Order_ID FK UNIQUE ,
Delivery_Date ,
Status
Tracking UNIQUE

)

Output :

The given ER diagram is successfully converted into a relational schema. All tables are created successfully with primary keys, foreign keys, NOT NULL, UNIQUE, ON DELETE CASCADE and ON inserted successfully and referential integrity violations are demonstrated.

5/8/26